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A PROJECT REPORT ON

"CLINISCAN: LUNG-ABNORMALITY DETECTION ON CHEST X-RAYS USING AI"

A Dissertation Submitted in partial fulfilment of the requirement for the degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE ENGINEERING

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I. INTRODUCTION

In recent years, chest X-ray imaging has become one of the most widely used diagnostic tools for detecting lung abnormalities such as pneumonia and other pulmonary infections. With the rapid advancement of deep learning, Convolutional Neural Networks (CNNs) have demonstrated significant potential in automating medical image classification tasks. Several studies have applied transfer learning models such as VGG19, EfficientNetB0, and Inception-V3 for pneumonia detection from chest X-ray images.

Furthermore, explainable AI techniques such as Grad-CAM and Grad-CAM++ have been employed to visualize model attention regions and enhance interpretability. However, most existing works primarily focus on binary classification and heatmap-based explanation without precise spatial localization or real-world system deployment.

The CliniScan project aims to design and implement an intelligent, end-to-end AI-powered diagnostic support system that integrates:

- Deep Learning – based classification
- YOLO based abnormality detection
- Grad CAM explainability
- Web – based Deployment

This integrated approach ensures not only high classification accuracy but also spatial interpretability and practical usability.

II. MOTIVATION BEHIND THE PROJECT WORK

Although several research works have demonstrated high classification accuracy in pneumonia detection, existing systems face key limitations:

- Lack of precise abnormality localization
- Limited deployment into real-time applications
- Absence of integrated diagnostic pipelines
- Focus primarily on binary classification

This motivates the need for a more comprehensive and practical AI solution that goes beyond

classification accuracy and provides:

- Localized disease detection
- Explainable predictions
- Real-time clinical prototype
- Integrated AI pipeline

CliniScan addresses these limitations by combining classification, detection, explainability, and deployment into a unified framework.

III. PROBLEM STATEMENT DEFINATION

Existing pneumonia detection systems predominantly rely on deep learning classification models that provide image-level predictions without accurate spatial localization. While Grad-CAM techniques highlight regions of interest, they do not provide bounding-box level precision required for clinical interpretation.

Moreover, most studies remain confined to experimental settings and lack real-time deployment interfaces.

The problem addressed in CliniScan is to develop an intelligent, integrated diagnostic system that:

- Classifies lung abnormalities
- Localizes affected regions using object detection
- Provides explainable visual outputs
- Enables real-time deployment through a web-based interface

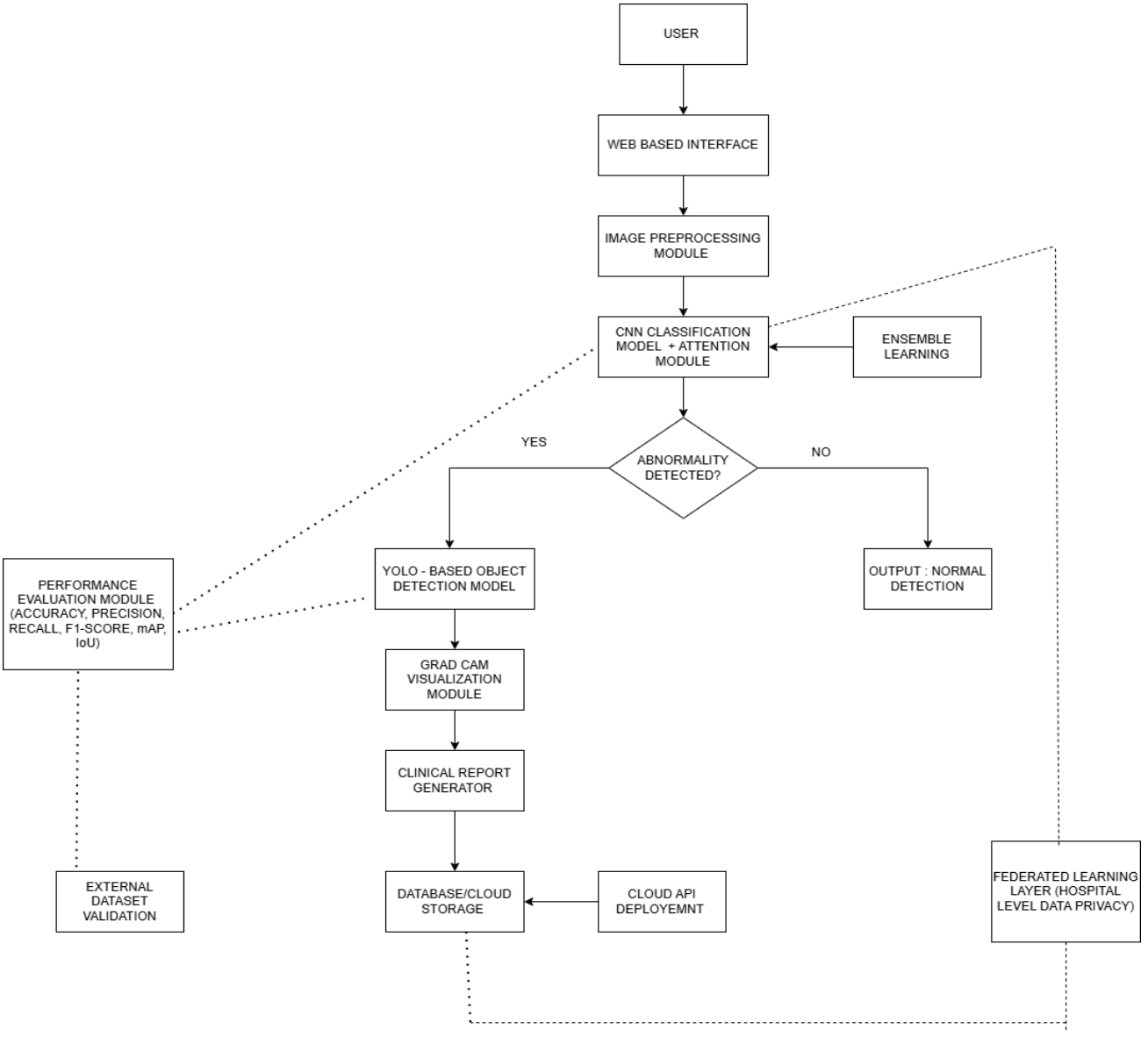
Such a system must enhance interpretability, clinical trust, and practical applicability beyond conventional classification models.

IV. STATEMENT OF OBJECTIVES

- To design a deep learning-based classification model using transfer learning for lung abnormality detection.
- To implement a YOLO-based object detection model for precise localization of abnormal lung regions.
- To integrate Grad-CAM visualization for improving interpretability and clinical trust
- To develop a web-based Streamlit prototype enabling real-time chest X-ray upload and analysis

- To evaluate system performance using accuracy, precision, recall, F1-score, IoU, and mAP.
- To compare CliniScan against existing classification-only models and demonstrate enhanced localization capability.

V. SYSTEM ARCHITECHTURE



VI. REFERENCES

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